

Park, Jung Hyun

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RESEARCH FOCUS & METHODS

Computational neuroscientist with integrated bioinformatics and wet-lab expertise in single-cell RNA-seq, transcriptomic analysis, multi-omics integration, and cross-species comparison. Ph.D. research focuses on glial lipid dysregulation in A β -tau co-pathology in Alzheimer's disease. Experience spans end-to-end transcriptomic workflows, Alzheimer's disease mouse models, and molecular/cellular biology techniques.

EDUCATION

Ph.D. in Neuroscience – Indiana University, Bloomington, IN	Aug 2022 – May 2026
- Advisors: Hui-Chen Lu and Jungsu Kim - Dissertation: <i>Decoding Aβ and Tau Co-Pathology in Alzheimer's Disease Mouse Models at Single-Cell Resolution</i>	
M.S., Kinesiology – Indiana University, Bloomington, IN	Aug 2020 – May 2022
- Advisor: Hannah J. Block - Thesis: <i>Specificity of Transcranial Magnetic Stimulation Effects on M1 and S1</i>	
Juris Doctor – Ewha Law School, Seoul, South Korea	Mar 2009 – Feb 2012
Bachelor of Law – Korea University, Seoul, South Korea	Mar 2004 – Feb 2009

LICENSURE & AFFILIATIONS

Patent Attorney , Korea Patent Attorneys Association	Feb 2015
Attorney-at-law , Korean Bar Association	May 2012

RESEARCH EXPERIENCE

Graduate Research Assistant – Indiana University, Indianapolis & Bloomington, IN Kim Lab (IU School of Medicine), Lu Lab (IU Bloomington)	Aug 2022 – Present
- Led experimental and computational transcriptomic studies of amyloid, tau, and combined pathology mouse models to identify cell-type-specific programs associated with Aβ-tau interaction and glial lipid dysregulation. - Conducted R/Python-based analyses of large-scale transcriptomic datasets, including cross-species alignment with human Alzheimer's disease signatures. - Developed and applied a targeted RNA-seq panel focused on lipid metabolism and immune pathways, enabling pathway-level resolution beyond whole-transcriptome approaches - Supported large-scale multiomic analyses integrating snRNA-seq and snATAC-seq in Aβ amyloidosis models. - Contributed to developing semi-automated single cell annotation pipelines, establishing reproducible analysis practices for RNA-seq datasets, and benchmarking nuclei isolation methods. - Generated publication-quality visualizations and manuscript-ready analyses to communicate biologically meaningful insights from high-dimensional omics data across experimental and computational collaborators.	
Graduate Research Assistant – Indiana University, Bloomington, IN Sensorimotor Neurophysiology Laboratory	Aug 2020 – May 2022
Conducted human neuromodulation (TMS) studies and analyzed behavioral and physiological outcomes	
Volunteer Research Assistant – Indiana University, Bloomington, IN Lai Laboratory – Trained in BAC recombination (Bacterial Artificial Chromosome engineering) Newman Memory Laboratory – Assisted with rat handling and optogenetic apparatus setup	Sep 2021 – Jul 2022 Nov 2019 – May 2020

PUBLICATIONS

- Park, J.H.**, ..., & Kim, J., (2026). Unique transcriptomic alterations in 5XFAD;PS19 mouse model identify glial lipid dysregulation and coordinated microglial–oligodendrocyte responses. *Alzheimer's & Dementia* 22.
- Kersey, H. N., ..., **Park, J. H.**, & Kim, J. (2026). Comparative analysis of nuclei isolation methods for brain single-nucleus RNA sequencing. *Cell Reports Methods*, Volume 6.
- Aciri, D. J., ..., **Park, J. H.**, ..., & Kim, J. (2026). Semi-automated annotation refinement accelerates cell type identification in brain spatial and single-cell studies. *bioRxiv*, 2026.2007.2030.741795.
- Park, J. H.**, ..., & Kim, J. (Work in progress) Restoring the statistical power failure and rigor of research using amyloid- β amyloidosis mouse models for Alzheimer's disease research.

- Owen T. Carmichael, ..., **Park, J. H.**, ..., Debora Melo Van Lent. (2025) Ultraprocessed foods and the aging brain: state of the science. *Annual Review of Nutrition*, Volume 46.
- Kim, B., ..., **Park, J. H.**, Mathias Jucker & Kim, J. (2024). Effects of SPI1-mediated transcriptome remodeling on Alzheimer's disease-related phenotypes in mouse models of A β amyloidosis. *Nature Communications*, 15(1), 3996.
- Yang, S., **Park, J. H.**, & Lu, H. C. (2023). Axonal energy metabolism, and the effects in aging and neurodegenerative diseases. *Molecular Neurodegeneration*, 18(1), 49.
- Park, J. H.**, Benson, R. F., Morgan, K. D., Matharu, R., & Block, H. J. (2023). Balance effects of tactile stimulation at the foot. *Human Movement Science*, 87, 103024.
- Park, J. H.** (2022). Reliability of theta burst stimulation as a neuromodulation tool. *Journal of Neurophysiology*, 127(6), 1532-1534.
- Park, J. H.** (2022). *Specificity of TMS Effects on M1 and S1* (Doctoral dissertation, [Bloomington, Ind.]: Indiana University).

TECHNICAL SKILLS

Mouse AD models | sc/snRNA-seq | Transcriptomics, Multiomics | Cross-species alignment | R, Python | HPC | GitHub

HONORS & AWARDS

- Three Minute Thesis (3MT) Competition Finalist – Indiana University 2026
- AGE Early Career Scholar Honorable Mention – American Aging Association 2025
- Gill Fellowship – Gill Center for Biomolecular Science, Indiana University 2023
- Travel Awards – Program of Neuroscience, Indiana University 2022, 2024, 2025
- Research Award – Graduate and Professional Student Government, Indiana University 2022
- Updyke/President's Challenge Fellowship – Indiana University 2021
- Sports Merit Medal – President of the Republic of Korea 2019
- Best Employee of the Year – PyeongChang Organizing Committee for the 2018 Olympic & Paralympic Winter Games 2016

POSTERS & PRESENTATIONS (First Author)

- *Unique transcriptomic alterations in combined amyloid and tau mouse models aligned with human Alzheimer's disease* – Indiana University Program of Neuroscience Poster Session, Bloomington, IN Jun 2025
- *Restoring the statistical power failure and rigor of research using amyloid- β amyloidosis mouse models for Alzheimer's disease* – Society for Neuroscience, Washington, DC Nov 2023
- *Specificity of TMS effects on the primary motor and sensory cortices* – Greater Indiana Chapter Society for Neuroscience Annual Meeting, Bloomington, IN Apr 2022

SERVICE & TEACHING

Early Career Policy Ambassador: Society for Neuroscience 2026

Reviewer: Molecular Neurodegeneration (2023), American Aging Association Early Career Scholar's Award (2025)

Guest Lecturer:

- Stark Medical Neuroscience Undergraduate Summer Research Program, Indiana University 2023 – 2024
- The Science Behind Brain Development and Function (Y488/P657), Indiana University 2023
- Annual Education Program for Government Officials, Korea Ministry of Government Legislation 2015 – 2017

Teaching Assistant: Advanced Techniques in Molecular Neuroscience Course, Cold Spring Harbor Laboratory 2022

WORKSHOPS

- Computational Techniques and Resources for Effective Translational Research in Alzheimer's Disease – The Jackson Laboratory May 2025
- Molecular Biology Summer Workshops – New England Biolabs Jul 2023
- Human Pluripotent Stem Cell Culture Workshop – STEMCELL Technologies Oct 2023

EARLIER PROFESSIONAL EXPERIENCE (LAW & GOVERNMENT)

Attorney-at-Law & Patent Attorney (South Korea) 2012–2018

- PyeongChang 2018 Organizing Committee: Legal Counsel (2014–2018); Director, Legal Affairs (2018)
- Embassy of Korea in Wellington, New Zealand: Research Officer (2013–2014)
- KY Law firm: Attorney-at-Law (2012–2013)